Two questions, simple answers

Group 2 · Building Age Classification · team briefing for the presentation and Q&A

1 · Why were 3 broken satellite scenes removed?

A **scene** is one Landsat photo of the whole area on one date. Each scene comes with a quality flag that marks, pixel by pixel, “good value” or “cloud”.

The problem

In 3 scenes the flag said “good” for almost every pixel, but the values were clearly wrong: nearly the whole image was extremely bright in the Blue band. That is the signature of haze, thin cloud, fog or snow that the cloud detector missed.

Scene	Satellite	Date	Note
LE07_L2SP_198023_20020417	Landsat 7	17 Apr 2002	
LT05_L2SP_199023_19961109	Landsat 5	9 Nov 1996	winter, low sun
LT05_L2SP_201032_19951207	Landsat 5	7 Dec 1995	winter, low sun

How we detected them (no labels used)

- A scene was flagged if its **typical (median) Blue value over all “good” pixels was above 12,000**, or **more than 5 % of its pixels were above 15,000**.
- A Blue value of 12,000 is about 13 % reflectance. Normal roofs and roads are well below that, so a whole scene sitting that high means the image is washed out.
- Only these 3 scenes, out of hundreds, crossed the threshold.

Why it matters

- Our strongest features look for a **sudden jump** in each pixel’s 42-year history: the “construction step”.
- One washed-out photo makes every pixel in that scene jump for that year, so the model would see **fake construction** across a whole area in 1995, 1996 or 2002.
- The starter notebook kept these photos; together with the other cleaning rules, removing them cut year-to-year noise by up to 46 %.

> **Say it like this: “Three satellite photos were secretly washed out by haze or cloud even though the quality flag said they were fine. They would have looked like fake construction events, so we removed them.”**

2 · What does “2 km spatial blocks, so neighbours cannot leak” mean?

The problem: neighbours are near-copies

- Buildings of the same age come in whole districts: **87 % of neighbouring pixels have the same age class**.
- Neighbouring pixels also have almost identical satellite histories, and some Madrid features average the surrounding pixels (windows up to 15 × 15 pixels = 450 m).

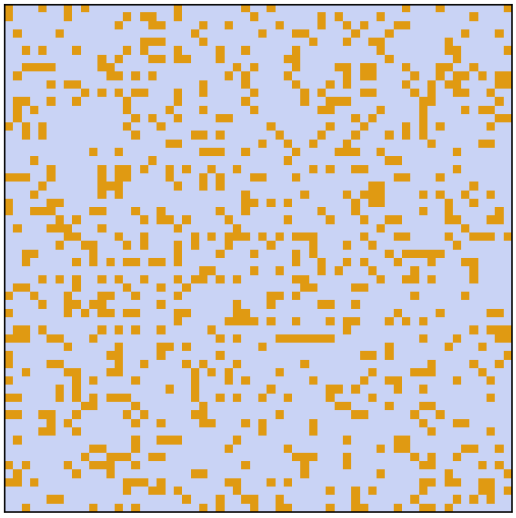
What goes wrong with a random split

- To test a model we hide some labelled pixels (test), train on the rest, and check the hidden ones.
- With a **random** split, a test pixel’s next-door neighbour is almost always in training. The model simply recognises the district, like a student who has seen the answers on the neighbour’s exam paper.
- The score looks great but is **inflated**: in Madrid a random split gives **0.736**.

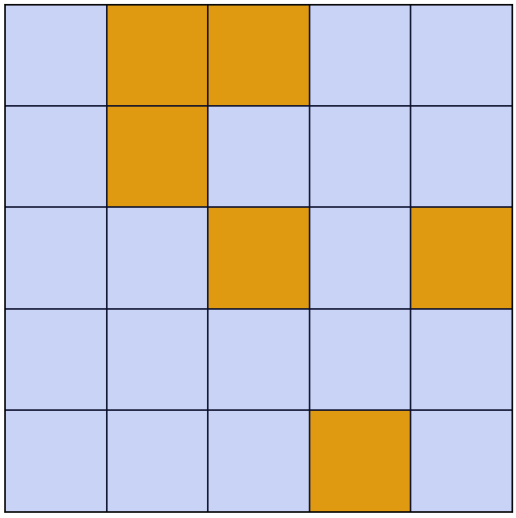
What we do instead

- Cut the map into squares of **67 × 67 pixels = 2 km × 2 km**.
- A whole square goes either to training or to testing, never both. Most test pixels are hundreds of metres to 2 km from any training pixel.
- The model must recognise age from the physics, not memorise the district. The honest Madrid score is **0.652**.

Random split → 0.736 (inflated)



2 km block split → 0.652 (honest)



Each small square = one 30 m pixel. Blue = training pixel, orange = test pixel. Right: each bold square = 2 km × 2 km block.

Why 2 km?

- Much larger than our widest neighbourhood feature (450 m), so no feature can carry information across a block edge.
- Roughly the size of a city district.
- The hidden test set may contain pixels from areas the model has never seen. The block score predicts that situation; the random score does not.

Confirmed in Amsterdam (experiment notebook 03)

- Labels taken only from one half of the 2 km blocks, scores measured only on the other half.
- Result: **0.65 → 0.75, average 0.711**, the same as the official protocol (0.712). Checked on two machines.
- Pixels more than 1 km from any label scored just as well (0.65 – 0.76), so our Amsterdam score does not come from neighbouring labels.

Say it like this: "We test on whole 2 km squares the model never saw, so it cannot cheat by copying neighbours. That is why our Madrid score is 0.65 and not the 0.74 a random split would give."